EVIDENCE MEMORANDUM | v6 — Deeper-investigation update

Sharp/Grove, Project Pandora/BIZARRE, and RF-Hearing Evidence Update

Version 6 incorporates the deeper-investigation findings on the SBIR AF93-026 / F41624-95-C-9007 program trail, the U.S. Air Force RF-hearing patent family, the Brunkan and MEDUSA prior-art leads, and the modern microwave-bioeffects and Havana-syndrome literature. It preserves the cautious counsel-facing posture of v5 and downgrades only those claims that the deeper record continues to leave under-corroborated.

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Document type	Working evidence memorandum / counsel briefing
Version	v6 — supersedes v5. Upgrades AF93-026 / F41624-93-C-9013 from "unverified mirror lead" to "authenticated-but-incomplete Air Force program trail" via the USAF FY2000 FOIA log corroboration of Phase II contract F41624-95-C-9007. Adds findings on USAF patents US6470214B1 and US6587729B2, the Brunkan US4877027A patent, the MEDUSA program reporting, the Foster/Garrett/Ziskin review, the National Academies 2020 report, the Bartholomew/Baloh skeptical review, and pulsed high-power RF literature. Maintains the v5 downgrade of the 1973 Sharp/Grove voice/digit claim to a single-source personal communication.
Subject	Joseph C. Sharp; H. Mark Grove; O. P. Gandhi; the WRAIR Department of Microwave Research; Project Pandora (1965–1970) and Project BIZARRE; the Justesen 1975 secondary report of a 1973 voice/digit demonstration; the Sharp/Grove/Gandhi 1974 IEEE acoustic-signals paper; later USAF RF-hearing patents (US6470214B1; US6587729B2); the SBIR AF93-026 / F41624-93-C-9013 Phase I and F41624-95-C-9007 Phase II program trail; Brunkan US4877027A; MEDUSA; modern weaponization caution.
Standard of proof	Cautious / source-backed. Distinguishes primary-document proof from peer-reviewed literature, secondary press, archived mirrors of dead federal abstract pages, and unauthenticated leads. Does not assert that mind control was proven, that any RF-hearing speech weapon was operationally deployed, or that listener-naive intelligible speech via RF was publicly demonstrated.
Intended use	Pre-litigation investigative reference; FOIA and archive planning; counsel review prior to any filing or public characterization.
Caveat	Not legal advice. Source provenance and admissibility must be independently verified by counsel of record before evidentiary use.

Standard-of-proof disclosure

This memorandum maintains careful evidentiary phrasing — supports an inference, corroborates official interest, evidence of a research environment, does not by itself prove. No primary-document 1973 task order tying Sharp and Grove to a classified speech-modulation tasking has been located. The 1979 Senate staff report[11] and the RAND/Koslov 1969 review[9] describe Pandora behavioral data as inconclusive or negative within reasonable scientific criteria; the National Security Archive's 2022 overview states there was no direct human radiation testing under PANDORA/BIZARRE[12]. The deeper investigation upgrades the Air Force SBIR program trail[27,28] and the USAF RF-hearing patent record[25,26] but does not establish operational deployment, listener-naive intelligible RF speech, or mind control[18,25,26].

1. Executive conclusion

Across the v6 record, four statements can be made with primary-document or officially-corroborated support, and three further statements remain explicitly limited.

Supported by primary or officially corroborated records

(a) Joseph C. Sharp and H. Mark Grove of the WRAIR Department of Microwave Research were principal participants in the Pandora-era microwave-bioeffects program. The 1979 Senate staff report names Maj. Joseph Sharp as the initial principal investigator of the Pandora primate work funded by ARPA at WRAIR[11]. The October 1966 JHU/APL operational-procedure document for the Pandora microwave test facility lists J. Sharp and H.M. Grove on its external distribution[4]; the April 21, 1969 Pandora meeting minutes record H. Mark Grove (WPAFB) reviewing USS Saratoga / Big Boy radar power-density measurements[7].

(b) The same Sharp/Grove pair, joined by O. P. Gandhi (Utah), published a 1974 IEEE paper (manuscript Oct. 25, 1973; revised Jan. 11, 1974) demonstrating audible clicks generated by 1500 MHz, 14-µs pulsed microwave energy in absorber materials[2]. The Foster & Finch 1974 Science paper established the underlying thermoacoustic mechanism[15]. Neither paper makes any claim of speech, voice, digit-word transmission, or mind control.

(c) The U.S. Air Force pursued RF-hearing speech communication seriously enough to obtain two patents — U.S. 6,470,214 B1 (filed Dec. 13, 1996; issued Oct. 22, 2002)[25] and the divisional U.S. 6,587,729 B2 (issued July 1, 2003)[26] — both assigned to the Department of the Air Force and now expired. The parent patent itself acknowledges that prior pulsed-carrier speech was distorted, was recognizable as speech only when the listener was preadvised of the content, and was unintelligible if the listener did not know the message in advance[25].

(d, new in v6) The Air Force SBIR program trail moves from "unverified activist mirror" to authenticated-but-incomplete. The USAF FY2000 FOIA log contains case 00-F-0013 (Margo P. Cherney) requesting "Copies of Communicating Via the Microwave Auditory Effect — DOD SBIR Contract #F41624-95-C-9007"[28]. That officially corroborates the existence of the Phase II contract referenced in the archived Phase I abstract[27]. Phase I details (AF93-026; control 93AL-185; F41624-93-C-9013; Brian Kohn / Science & Engineering Associates; $37,806; 17 May–17 Dec 1993) still rely materially on archived mirrors of a defunct federal SBIR abstract page[27].

Limits and counter-evidence the record imposes

(e) The single source linking Sharp and Grove to a speech / digit-word demonstration is Justesen's 1975 American Psychologist article, in which the relevant claim is supported by Note 2: "Sharp, J. C., & Grove, M. Personal communication, September 28, 1973."[1] This is a published secondary account citing a personal communication. No corresponding Sharp/Grove technical report has been located in the Glaser bibliography[3], the WRAIR FY1983 Annual Report[21], the WRAIR/USAMRD 2004 microwave laboratory contract report[22], the WRAIR archival finding aid[19], the WRAIR 1973–1974 annual report in the public record[37], or Justesen's own 1979 follow-up paper[36], which does not repeat the voice/digit anecdote.

(f) The Pandora behavioral record is, by its own primary documents, inconclusive or negative. The November 1969 RAND/Koslov review found the data did not present evidence of behavioral change due to the special signal within reasonable scientific criteria, and attributed apparent effects to systematic error and equipment malfunction[9]. The 1979 Senate report repeats and extends that conclusion: McIlwain found no significant effects; Kubis/SAC reviewers called Sharp's work exploratory with weak baselines, no controls, and equipment failures; the laboratory human studies were never carried out due to DOD informed-consent concerns[11]. The 2022 NSArchive overview confirms no direct human radiation testing under PANDORA/BIZARRE was performed before the program closed[12].

(g) The patents and the SBIR program trail document official interest and a claimed signal-processing solution; they do not document field-deployable, listener-naive intelligible speech delivered by microwave or RF energy. The Foster/Garrett/Ziskin modern review concludes that practical weaponization with currently known systems is unlikely or operationally implausible[18]. The National Academies 2020 report found acute signs and symptoms in some State Department personnel most consistent among the considered mechanisms with directed, pulsed RF energy[32,33], while a later skeptical review by Bartholomew & Baloh emphasizes the 2023 U.S. intelligence-community judgment that a foreign sonic or microwave device is highly unlikely to explain reported cases[34]. Recent pulsed high-power RF bioeffects literature keeps biological plausibility open without validating operational voice-to-skull speech[35].

> Bottom line. The deeper investigation strengthens the middle of the chain — there was a real Air Force research-and-patent trail on RF-hearing speech communication in the 1990s, and the SBIR Phase II contract is officially corroborated by a USAF FOIA log. The deeper investigation does not rescue the strongest public claims. Sharp/Grove 1973 remains uncorroborated beyond Justesen's personal communication; PANDORA/BIZARRE does not prove mind control; patents do not prove deployment; and no open primary source reviewed here establishes reliable operational listener-naive RF speech communication.

2. Evidence ledger

The ledger separates evidence by provenance class. Strength labels denote the evidentiary category, not the truth of any underlying claim. Primary = original government or contemporaneous published record. Peer-review = peer-reviewed or technical literature. Authenticated-incomplete = an officially corroborated program element with key technical deliverables not yet in the public file. Secondary = press, FOIA-derived narrative, or summary of an underlying record. Mirror = archived copy of a now-defunct federal abstract page, treated as a lead pending official authentication. Negative = a deliberate search that did not find what was sought.

Class	Item	Finding	Sources
PRIMARY	1979 Senate staff report	Names Maj. Joseph Sharp as initial PI of the Pandora primate work; documents 4–5 mW/cm² special signal; lab human studies never carried out due to DOD informed-consent concerns; classified records destroyed Sept. 1973 and June 29, 1978; Pandora terminated March 20, 1970.	11
PRIMARY	1966 JHU/APL Operational Procedure for Pandora Microwave Test Facility	October 1966 document for the WRAIR/Forest Glen S-band facility; external distribution list includes J. Sharp and H.M. Grove.	4
PRIMARY	Apr. 21, 1969 Pandora meeting minutes	H. Mark Grove (WPAFB) reviewed USS Saratoga / Big Boy radar measurements; ≤ 1 mW/cm² at 80–90% radar operation; no significant differences; WRAIR human studies proposed but not finally approved.	7
PRIMARY	May 12, 1969 Pandora human-protocol minutes	Preliminary classified WRAIR draft for 8 subjects, 90 days, 4.5–5 mW/cm². Planning, not execution.	8
PRIMARY	1965 ARPA / Cesaro initiating memorandum	Establishes Pandora's mandate around radiation effects on humans tied to the Moscow Embassy / TUMS signal; Mark Grove (WPAFB Avionics Lab) as ARPA monitor/coordinator. No speech-transmission reference.	5
PRIMARY	Project BIZARRE progress report	Pandora/TUMS context; modulated test levels; behavioral-control / weapon-application language. Important context, not proof of weapon or speech system.	6
PRIMARY	RAND/Koslov 1969 review	To Lukasik (DARPA): Pandora data did not present evidence of behavioral change due to the special signal within reasonable scientific criteria; behavioral changes attributable to systematic error or equipment malfunction.	9
PRIMARY	1977 DARPA letter to Senator Magnuson	Pandora terminated March 1970; assets transferred to WRAIR; total cost $4,615,000; raw-data disposition unknown.	10
PRIMARY	NSArchive 2022 "Moscow Signals Declassified" overview	PANDORA/BIZARRE failed to prove the Soviet-beam degradation hypothesis; corrected Moscow power ≤ 0.05 mW/cm²; no direct human radiation testing under PANDORA/BIZARRE.	12
PRIMARY	U.S. Patent 6,470,214 B1 (USAF; O'Loughlin & Loree)	Method and device for implementing the RF hearing effect. Suppressed-carrier modulation, audio preprocessor, square-root processor. Patent text states pulsed-carrier modulation distorted speech as experimentally confirmed and that intelligibility could depend on listener pre-advice.	25

Class	Item	Finding	Sources
PRIMARY	U.S. Patent 6,587,729 B2 (divisional)	Apparatus for audibly communicating speech using the RF hearing effect. Inherits the parent's prior-art acknowledgment that, if the listener did not know the content, the audio signal was unintelligible in the prior experimental context.	26
AUTH-INCOMPLETE	USAF FY2000 FOIA log — case 00-F-0013	Cherney request for "Communicating Via the Microwave Auditory Effect — DOD SBIR Contract #F41624-95-C-9007." Officially corroborates the Phase II contract; the technical results themselves are not in the public file reviewed here.	28
AUTH-INCOMPLETE	SBIR Phase I AF93-026 / F41624-93-C-9013 (Kohn / SEA, 1993)	Phase I details (control 93AL-185; $37,806; 17 May–17 Dec 1993; reportedly concluding voice communications via the microwave auditory effect are highly feasible) survive principally on an archived mirror of a defunct federal SBIR abstract page. Materially corroborated by the Phase II FOIA-log entry above and the later USAF patent family.	27, 28, 25, 26
PEER-REVIEW	Sharp, Grove, Gandhi 1974 (IEEE Trans. MTT)	1500 MHz, 14-µs pulses, 3 pps; acoustic signals/clicks from absorbers; delays consistent with sound propagation. Confirms acoustic signals in absorbers. Does not mention voice, speech, or Pandora.	2
PEER-REVIEW	Foster & Finch 1974 (Science)	Thermoacoustic mechanism for microwave hearing/clicks. Mechanism evidence; not speech evidence.	15
PEER-REVIEW	Guy, Chou, Lin, Christensen 1975 (Annals NYAS)	Microwave-induced acoustic effects in mammalian auditory systems. Technical context.	16
PEER-REVIEW	Lin 1978 (Microwave Auditory Effects and Applications)	Standard reference textbook on the microwave hearing mechanism.	17
PEER-REVIEW	Foster, Garrett, Ziskin (modern review)	Concludes that practical weaponization with currently known systems is unlikely or operationally implausible; classified high-power RF uncertainties prevent absolute closure.	18
PEER-REVIEW	National Academies 2020 (RF mechanisms chapter)	Acute signs and symptoms reported by State Department personnel most consistent among considered mechanisms with directed, pulsed RF energy; committee had limited information; no source identification.	32, 33
PEER-REVIEW	Bartholomew & Baloh skeptical review	Cites 2023 U.S. IC assessment that a foreign sonic or microwave device is highly unlikely to explain Havana cases; criticizes earlier studies for over-reading weak clinical data.	34
PEER-REVIEW	Yaghmazadeh — pulsed high-power RF bioeffects	Biological plausibility for some neurological effects; does not validate operational voice-to-skull speech.	35
SECONDARY	Justesen 1975 "Microwaves and Behavior" (Note 2)	Speech/digit-word claim sourced explicitly to Note 2: "Sharp, J. C., & Grove, M. Personal communication, September 28, 1973." Note 1: Guy, A. W., personal communication, October 15, 1973. Single-source, secondary.	1
SECONDARY	Justesen 1979 follow-up	Does not repeat the Sharp/Grove voice/digit anecdote — a negative continuity signal from the original author.	36
SECONDARY	CIA Reading Room — Washington Post clipping (May 10, 1972)	"'BRAINWASH' ATTEMPT BY RUSSIANS?" Press clipping retained in CIA files. Press corroboration only.	13

Class	Item	Finding	Sources
SECONDARY	Foreign Policy 2017 narrative	Sharp said to have left in 1968. Use cautiously; the 1979 Senate report's chronology should govern.	14
SECONDARY	MEDUSA reporting (New Scientist; Wired, 2007–08)	Sierra Nevada "Mob Excess Deterrent Using Silent Audio"; Phase I hardware design; James Lin warns about power and possible neural damage; no field deployment documented.	30, 31
MIRROR	GBPPR archived SBIR Phase I abstract	Archived copy of a defunct federal SBIR/EPA abstract page. Source for the Phase I narrative details for AF93-026.	27
PRIOR-ART	U.S. Patent 4,877,027 — Brunkan "Hearing system"	Microwave-induced hearing using 100 MHz–10 GHz radiation, bursts of pulses, audio modulation. Patent claims and prior-art evidence; not deployment proof.	29
NEGATIVE	Glaser bibliography — no "Sharp & Grove Note 2"	Closest entries: Sharp & Papeziello 1971 (rat thymidine uptake); MR0094 Hawkins/Grove/Heiple/Schrot (Mar. 1973, 32 pp.); Gandhi entries; Justesen 1975/1978; broader microwave-hearing literature.	3
NEGATIVE	WRAIR FY1983 annual report	Work Unit 214 (Millimeter Wave Biophysics); Work Unit 041 (Microwave Radiation Hazards); no Sharp/Grove/Gandhi/Justesen/Pandora/RF-hearing hits.	21
NEGATIVE	WRAIR/USAMRD 2004 contract report	Microwave Bioeffects Branch (Brooks City-Base) under DAMD17-94-C-4069 / DAMD17-94-E-0001; mentions microwave hearing review; no Sharp/Grove/Pandora/speech.	22
NEGATIVE	PubMed WRAIR Department of Microwave Research continuity	Authorship persists into the 1990s in behavioral-timing/microwave papers; no Sharp/Grove/speech.	23, 24
NEGATIVE	WRAIR 1973–74 annual report (DTIC ADA009338)	Establishes the relevant WRAIR microwave-research environment for Sharp/Grove; the public copy reviewed contains no matching voice/digit experiment.	37

3. New v6 findings and corrections

3.1 SBIR program trail upgraded — official FOIA-log corroboration

The deeper investigation's strongest v6 addition is the discovery that the U.S. Air Force FY2000 FOIA log (released and redacted, hosted by The Black Vault) records case 00-F-0013, requester Margo P. Cherney, subject "Copies of Communicating Via the Microwave Auditory Effect — Awarding Agency: DOD SBIR Contract #F41624-95-C-9007"[28]. That log is an official Air Force record; it authenticates the existence of the Phase II SBIR contract that was previously known only through archived mirrors. The Phase II contract is therefore upgraded from "unverified mirror lead" to "officially corroborated, technical results not in the public file reviewed here."

Phase I details — AF93-026, control 93AL-185, contract F41624-93-C-9013, awardee Brian Kohn / Science & Engineering Associates, $37,806, 17 May to 17 December 1993, with the abstract's high-feasibility conclusion — still rest principally on the GBPPR-hosted archived copy of a now-defunct federal SBIR/EPA abstract page[27]. Those Phase I details are now materially corroborated by the official Phase II FOIA-log entry[28] and by the contemporaneous USAF RF-hearing patent family[25,26], but the underlying Armstrong Laboratory / OEDR feasibility study referenced in the Phase I abstract has not been located in open DTIC or official records. The evidentiary posture is therefore real program, incomplete public file — not hoax, and not proven operational system.

3.2 USAF RF-hearing patent family — authentic, narrowly read

U.S. Patent 6,470,214 B1 was filed December 13, 1996 and issued October 22, 2002, with James P. O'Loughlin and Diana L. Loree as inventors and the United States Department of the Air Force as assignee[25]. The divisional U.S. Patent 6,587,729 B2 issued July 1, 2003 and covers an apparatus for audibly communicating speech using the RF-hearing effect[26].

The most legally and scientifically important sentence in the parent patent is the prior-art acknowledgment that earlier pulsed-carrier RF-hearing speech could be recognized as speech only when the listener was preadvised of the content, and was unintelligible if the listener did not know the message in advance[25]. The patent then claims a signal-processing solution — suppressed-carrier modulation, an audio preprocessor, and a square-root processor — intended to render speech intelligible through the RF-hearing effect[25,26]. That claim language is not equivalent to publicly available listener-naive test scores or operational deployment evidence: a patent claim is not a public test data set.

Both patents have expired, which is consistent with non-transition, non-maintenance, or loss of continuing patent-protection value, but expiration alone cannot prove technical failure[25,26].

3.3 1973 Sharp/Grove voice/digit claim — downgrade preserved

The deeper investigation does not strengthen the 1973 Sharp/Grove voice/digit anecdote. Justesen's 1975 article specifically describes Sharp and Grove using zero-crossing-triggered microwave pulses derived from voice recordings of digit words and reports they could hear and distinguish the words, but the supporting reference is Note 2: "Sharp, J. C., & Grove, M. Personal communication, September 28, 1973"[1]. The contemporaneous Sharp/Grove/Gandhi IEEE paper documents 1500 MHz, 14-µs pulses producing acoustic signals in absorber materials — clicks, not voice; absorber, not human listener[2]. The two records are materially different in subject matter.

v6 negative searches confirm the v5 result: no "Sharp and Grove Note 2"-style underlying record was located in the Glaser bibliography[3], the WRAIR FY1983 annual report[21], the WRAIR/USAMRD 2004 microwave laboratory contract report[22], the WRAIR archival finding aid[19], the WRAIR 1973–1974 annual report in the public record[37], or Justesen's own 1979 follow-up paper[36]. Justesen 1979's silence on the voice/digit anecdote is a negative signal because the original author did not later supply an updated citation or a stronger source.

3.4 Brunkan US4877027A and MEDUSA — claim/prototype/prior-art only

The Brunkan "Hearing system" patent (US4877027A) claims microwave-induced hearing using 100 MHz to 10 GHz radiation, bursts of pulses, and audio modulation[29]. The patent itself states preferences and claimed mechanisms; it does not document a fielded system. It is useful as prior-art and claims evidence in the RF-hearing speech lineage, not as proof of deployed capability.

MEDUSA — Sierra Nevada Corporation's proposed "Mob Excess Deterrent Using Silent Audio" — is documented in mainstream reporting that describes prototype-oriented Phase I work, hardware design and construction, and experimental indications of the microwave auditory effect, while quoting James Lin warning about power-level feasibility and possible neural damage[30,31]. The reporting does not document field deployment. The MEDUSA record is prototype/prior-art evidence, not deployment proof.

3.5 Modern feasibility, biological plausibility, and Havana-syndrome context

The 2021 Foster, Garrett, & Ziskin review concludes that microwave auditory sensations can be induced, but weaponization as a practical harassment or harm system appears unlikely for reasons of effect size and practicality, while classified high-power RF uncertainties prevent absolute closure[18]. The same review notes that existing microwave systems could produce unexpected or distracting auditory sensations but would be large or obvious, and it does not support the conclusion that such systems caused Havana incidents[18].

The National Academies of Sciences, Engineering, and Medicine 2020 report found many distinctive acute signs and symptoms reported by State Department personnel most consistent among the considered mechanisms with directed, pulsed RF energy, while emphasizing that the committee had limited information and was not positioned to identify a source or exposure circumstances[32,33]. A later skeptical review by Bartholomew & Baloh argues that U.S. intelligence agencies judged a foreign sonic or microwave device "highly unlikely" in 2023 and criticizes earlier Havana-syndrome studies for over-reading weak clinical data[34]. Recent pulsed high-power RF literature keeps biological plausibility open without validating operational voice-to-skull speech communication[35].

Important conceptual distinction. Causing a biological or perceptual effect is easier to establish than delivering reliable, private, intelligible, listener-naive speech at useful range and power levels[18,35]. The biological-plausibility question and the operational-communication question should not be conflated.

4. RF-hearing physics and the patent record

4.1 Thermoacoustic mechanism

The Foster & Finch 1974 Science paper established that microwave hearing arises from rapid thermoelastic expansion in head tissue caused by absorbed pulsed-microwave energy[15]. The Sharp/Grove/Gandhi 1974 IEEE paper is consistent with this physics: 1500 MHz, 14-µs pulses at 3 pps produced acoustic signals and clicks from absorbers, with delays consistent with sound propagation[2]. The Guy/Chou/Lin/Christensen 1975 Annals NYAS paper extended the literature on microwave-induced acoustic effects in mammalian auditory systems[16], and Lin's 1978 monograph remains the standard reference[17].

None of these foundational papers describe intelligible speech transmission. The leap from clicks (or acoustic signals in absorber materials) to listener-naive intelligible speech is not established in the open peer-reviewed record[18].

4.2 USAF patents — primary record of an RF-hearing speech apparatus

U.S. Patent 6,470,214 B1 (O'Loughlin & Loree; assignee U.S. Department of the Air Force; filed Dec. 13, 1996; issued Oct. 22, 2002)[25] describes a method and device using suppressed-carrier modulation, an audio preprocessor, and a square-root processor to transmit RF to a subject's head so that the thermal-acoustic RF-hearing effect renders intelligible speech. The patent text states that pulsed-carrier modulation distorted speech as experimentally confirmed; the patent's prior-art discussion explicitly notes that, when the listener was not preadvised of the content, the audio signal was unintelligible[25]. U.S. Patent 6,587,729 B2 (filed Apr. 24, 2002; issued July 1, 2003) is the divisional, directed to apparatus claims for audibly communicating speech using the RF-hearing effect[26].

> How v6 reads these patents. They are primary evidence that the U.S. Air Force pursued an RF-hearing speech apparatus in the 1990s and early 2000s and obtained granted claims to a signal-processing solution. They are not evidence that the apparatus produced field-deployable, listener-naive intelligible speech: the patent documents themselves describe distortion under pulsed-carrier modulation and intelligibility dependent on prior knowledge of content, and the patent record is silent on operational test scores at useful range and power.

4.3 SBIR AF93-026 / F41624-93-C-9013 — Phase I

The archived Phase I abstract describes a 1993 Science & Engineering Associates award led by Brian Kohn under solicitation AF93-026, control 93AL-185, contract F41624-93-C-9013, to investigate communication using the microwave auditory effect and to extend an Armstrong Laboratory / OEDR feasibility study; period of performance 17 May to 17 December 1993; total $37,806; abstract reportedly concludes voice communications via the microwave auditory effect are highly feasible[27]. The Armstrong Laboratory / OEDR feasibility study has not been located in open DTIC or official records during this investigation. The Phase I narrative is therefore an authenticated-but-incomplete program lead — corroborated by the Phase II FOIA-log entry below, but not yet supported by an official primary copy of the Phase I final report.

4.4 SBIR F41624-95-C-9007 — Phase II (officially corroborated)

The USAF FY2000 FOIA log records case 00-F-0013 (Margo P. Cherney) requesting "Copies of Communicating Via the Microwave Auditory Effect — Awarding Agency: DOD SBIR Contract #F41624-95-C-9007"[28]. This is an official Air Force document. It authenticates the existence of the Phase II contract by name and contract number. It does not, by itself, place any technical Phase II deliverable in the public file. The next decisive record would be the Phase II final report and segregable interim deliverables, any Armstrong Laboratory / OEDR feasibility report, and any additional FOIA releases tied to case 00-F-0013.

4.5 Brunkan and MEDUSA — prior-art and prototype evidence

The Brunkan U.S. 4,877,027 patent claims microwave-induced hearing using 100 MHz to 10 GHz radiation, bursts of pulses, and audio modulation — useful as prior-art and claims evidence in this lineage, not as deployment proof[29]. The MEDUSA reporting documents prototype-oriented Phase I work and expert concern about power levels and safety; field deployment is not documented[30,31].

5. Overclaim risk register

Following the legal/risk-assessment framework, the table below scores the principal overclaim risks the user should avoid when characterizing the Sharp/Grove, Pandora/BIZARRE, and RF-hearing record publicly. Severity indicates the magnitude of damage to credibility or to a counsel's evidentiary position if the overclaim is repeated; likelihood indicates how easy it is to slip into the overclaim given how the literature is commonly summarized.

Overclaim	Severity	Likelihood	Score	Why this is a risk	Sources
Asserting that mind control was proven	Critical (5)	Possible (3)	15 — High	RAND/Koslov 1969 and the 1979 Senate report characterize Pandora behavioral evidence as inconclusive or negative; NSArchive 2022 says no direct human radiation testing under PANDORA/BIZARRE was performed.	9, 11, 12
Asserting an RF-hearing speech weapon was operationally deployed	Critical (5)	Possible (3)	15 — High	USAF patents describe an apparatus in the patent record only, with documented distortion and listener-pre-advice limitations. Foster/Garrett/Ziskin call practical weaponization with known systems unlikely.	18, 25, 26
Asserting listener-naive intelligible RF speech was publicly demonstrated	Critical (5)	Possible (3)	15 — High	The parent USAF patent itself acknowledges that prior pulsed-carrier speech was unintelligible if the listener was not preadvised; no public listener-naive test data has been located.	25, 26, 18
Asserting AF93-026 was a hoax because it appears principally on activist mirror sites	High (4)	Likely (4)	16 — Critical	The Phase II contract F41624-95-C-9007 is officially corroborated by the USAF FY2000 FOIA log. Treating the Phase I trail as a hoax misreads the deeper record.	27, 28
Asserting the SBIR or patent record proves field deployment	High (4)	Likely (4)	16 — Critical	An officially corroborated contract and an issued patent prove official interest and a claimed solution; they do not prove deployment, transition, or operational performance.	25, 26, 27, 28
Asserting a direct 1973 ARPA tasking to Sharp & Grove for speech	High (4)	Possible (3)	12 — High	No such 1973 task order has been located. Justesen Note 2 is a personal communication of Sept. 28, 1973; Sharp/Grove/Gandhi 1974 is silent on speech.	1, 2
Quoting Justesen's voice/digit anecdote as primary evidence	High (4)	Likely (4)	16 — Critical	Justesen 1975 is a published secondary account citing a personal communication; Glaser bibliography contains no "Sharp and Grove Note 2" entry; Justesen 1979 does not repeat it.	1, 3, 36
Citing Havana-syndrome reporting as proof of microwave voice weapons	High (4)	Likely (4)	16 — Critical	The National Academies report discussed directed pulsed RF as plausible for some symptoms; later reviews and the 2023 IC assessment dispute a device explanation. The literature does not establish voice-to-skull deployment.	32, 33, 34, 18

Overclaim	Severity	Likelihood	Score	Why this is a risk	Sources
Treating Pandora special-signal data as transferring to embassy conditions	High (4)	Possible (3)	12 — High	Corrected Moscow signal ≤ 0.05 mW/cm²; WRAIR special signal was 4–5 mW/cm² — orders of magnitude above corrected ambient; primate exposure conditions do not automatically transfer.	11, 12
Treating mindjustice.org / mirror-only sources as primary	Moderate (3)	Likely (4)	12 — High	Such mirrors are unauthenticated; obtain SBIR / USPTO file-wrapper or FOIA-released originals before quoting. The GBPPR mirror should be paired with the FOIA-log entry whenever cited.	27, 28
Citing Foreign Policy 2017 chronology as primary	Moderate (3)	Likely (4)	12 — High	Secondary press narrative; the 1979 Senate staff report and Pandora minutes should govern Sharp/McIlwain timing.	11, 14
Treating the 1972 Washington Post clipping as CIA analysis	Moderate (3)	Possible (3)	9 — Medium	The CIA Reading Room item is a press clipping retained in CIA files, not a CIA-authored intelligence assessment.	13

6. Recommended FOIA, USPTO, and archive next steps

Each request below ties to a specific corroborating record. Decisive next records, in priority order: (1) the Phase II SBIR final and interim deliverables for F41624-95-C-9007; (2) the Armstrong Laboratory / OEDR feasibility report referenced in the Phase I abstract; (3) the USPTO file wrappers for the O'Loughlin / Loree patents; (4) WRAIR archival folders for the Pandora period and 1973–1975 Department of Microwave Research correspondence.

Request target	Why this matters	Sources
AFRL FOIA — F41624-95-C-9007 (Phase II) interim and final deliverables, with segregable portions	The official FY2000 FOIA log authenticates the Phase II contract; the technical results are the missing decisive record.	28
AFRL / former Armstrong Laboratory — OEDR feasibility report referenced by the Phase I abstract	Pre-SBIR bridge between RF-hearing feasibility and the 1993 award; not located in open DTIC or official web records during this investigation.	27
DTIC — full-text search for AF93-026, F41624-93-C-9013, F41624-95-C-9007, Brian Kohn final reports, SEA	Authenticate the SBIR mirror against an official DTIC technical report; surface decisive technical-report deliverables.	27, 28
USPTO — file wrappers for U.S. 6,470,214 B1 (US08/766,687) and U.S. 6,587,729 B2 (US10/131,626)	Office actions, examiner correspondence, declarations, and any incorporation of unpublished test data behind the patent claims.	25, 26
WRAIR archive — Folder 128 "Project Pandora 1969–1970" (Series I)	Direct primary-document folder for the Pandora minutes period and Sharp's monkey-exposure interpretation.	11, 19, 20
WRAIR archive — Folders 112–115 (Microwave Research Program Review, 1981; vols. I–III) and Microwave Research Program 1982/1991	Post-Pandora retrospective program reviews; potential references back to 1965–1970 work and Sharp/Grove personnel files.	19
WRAIR archive — Series II Folder 26 "Microwave Research, Department of, 1975"	Closest-dated departmental folder to the Justesen 1975 article and the Sharp/Grove/Gandhi 1974 IEEE publication.	1, 2, 19
WRAIR archive — Department of Microwave Research correspondence and personal-communication files, 1973	Ask explicitly for any Sharp/Grove correspondence on or near September 28, 1973 (Justesen Note 2 date) or October 15, 1973 (Justesen Note 1 date).	1, 19, 20
WRAIR archive — directories/rosters/org manuals for 1969, 1971, 1973, 1975	Verify Department of Microwave Research staffing for Sharp and Grove across the Pandora and Justesen 1975 periods.	11, 19
ARPA / DARPA records — ARPA Order series for Pandora 1965–1970	Initial 1965 ARPA tasking, monitor designations, and any speech-modulation or auditory-bioeffects sub-task.	5, 10
NARA / presidential libraries — Pandora-period material (RG 330 DOD, RG 112 Surgeon General, Lukasik / DARPA Director collections)	NSArchive 2022 cites NARA and presidential libraries as document sources; consider follow-on FOIA at these record groups.	12
USAMRDC / WRAIR Brooks City-Base Microwave Bioeffects Branch	2004 contract report shows continuity into Brooks; FOIA the program records and any RF-hearing review file.	22
DTIC restricted holdings — Sharp, Grove, Gandhi, O'Loughlin, Loree, Kohn, SEA, Armstrong/OEDR	Open-web and public DTIC searches have not surfaced the decisive technical reports.	27, 28, 37

Suggested keyword set for archive and FOIA queries

Sharp; Grove; Gandhi; Department of Microwave Research; "receiverless"; "voice-modulated"; "RF hearing"; "microwave hearing"; "Note 2"; "personal communication"; Pandora; BIZARRE; TUMS; "special signal"; AF93-026; F41624-93-C-9013; F41624-95-C-9007; O'Loughlin; Loree; Kohn; Science & Engineering Associates; Armstrong Laboratory; OEDR; Brunkan; MEDUSA; Sierra Nevada Corporation; case 00-F-0013.

WRAIR archive access

WRAIR Library / Archival Collection access: in-person review by U.S. citizens, by appointment, at the Gorgas Memorial Library, 503 Robert Grant Avenue, Silver Spring, MD 20910; telephone 301-319-9555[20].

7. Master source list

All numbered footnote markers throughout this memorandum reference the list below. Every URL is clickable. Authentication status reflects what v6 was able to verify at the primary level.

#	Source	URL
1	Justesen, D.R. — "Microwaves and Behavior," American Psychologist (1975); Note 1 (Guy, A.W., personal communication, Oct. 15, 1973) and Note 2 (Sharp, J.C., & Grove, M., personal communication, Sept. 28, 1973).	https://zoryglaser.com/wp-content/uploads/2020/05/MICROWAVES-AND-BEHAVIOR.pdf
2	Sharp, J.C., Grove, H.M., Gandhi, O.P. — "Generation of Acoustic Signals by Pulsed Microwave Energy," IEEE Trans. Microwave Theory Tech. (manuscript Oct. 25, 1973; revised Jan. 11, 1974).	https://zoryglaser.com/wp-content/uploads/2020/05/GENERATION-OF-ACOUSTIC-SIGNALS-BY-PULSED-MICROWAVE-ENERGY.pdf
3	Glaser, Z.R. — Bibliography of Reported Biological Phenomena ('Effects') and Clinical Manifestations Attributed to Microwave and Radio-Frequency Radiation (NMRI / Glaser archive).	https://zoryglaser.com/wp-content/uploads/2020/05/BIBLIOGRAPHY-FOR-RADIOFREQUENCY-AND-MICROWAVE-RADIATION.pdf
4	OSD FOIA Reading Room — "Operational Procedure for Project Pandora Microwave Test Facility," E.V. Byron, JHU/APL (Oct. 1966); WRAIR/Forest Glen S-band anechoic facility; distribution list includes J. Sharp and H.M. Grove.	https://www.esd.whs.mil/Portals/54/Documents/FOID/Reading%20Room/Other/Operational_Procedure_For_Project_Pandora.pdf
5	ARPA/Cesaro 1965 initiating memorandum — Project Pandora origin; Mark Grove (WPAFB Avionics Laboratory) as ARPA monitor/coordinator. National Security Archive scan.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/03.pdf
6	Project BIZARRE progress report — Pandora/TUMS context; modulated test levels; behavioral-control / weapon-application language. NSArchive scan.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/08.pdf
7	April 21, 1969 Pandora meeting minutes — H. Mark Grove (WPAFB) reviewed USS Saratoga / Big Boy radar measurements; ≤ 1 mW/cm² at 80–90% radar operation; no significant differences.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/10.pdf
8	May 12, 1969 Pandora human-protocol minutes — preliminary classified WRAIR draft for 8 subjects, 90 days, 4.5–5 mW/cm²; planning, not execution.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/11.pdf
9	Koslov, S. (RAND) — "Review of Project Pandora Experiments," Nov. 4, 1969, to Stephen Lukasik (DARPA): no behavioral change demonstrated within reasonable scientific criteria.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/12.pdf
10	DARPA letter (1977) to Senator Magnuson — Pandora terminated March 1970; assets transferred to WRAIR; total cost $4,615,000.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/14.pdf
11	U.S. Senate, 1979 staff report — Pandora initiated early 1965 via ARPA funding to WRAIR; principal investigator Maj. Joseph Sharp; lab human studies never carried out (DOD informed-consent concerns); classified records destroyed Sept. 1973 and June 29, 1978.	https://nsarchive.gwu.edu/sites/default/files/documents/rhwvuj-tzy4b/15.pdf

#	Source	URL
12	National Security Archive — "Moscow Signals Declassified: Microwave Diplomacy" (Sept. 13, 2022): PANDORA/BIZARRE failed to prove Soviet-beam degradation hypothesis; corrected Moscow power ≤ 0.05 mW/cm²; no direct human radiation testing under PANDORA/BIZARRE.	https://nsarchive.gwu.edu/briefing-book/intelligence-russia-programs/2022-09-13/moscow-signals-declassified-microwave
13	CIA Reading Room — Washington Post clipping, "'BRAINWASH' ATTEMPT BY RUSSIANS?" (May 10, 1972). Press clipping in CIA files; not CIA-authored analysis.	https://www.cia.gov/readingroom/document/cia-rdp80-01601r000300340041-0
14	Foreign Policy — "The Secret History of Diplomats and Invisible Weapons" (Aug. 25, 2017). Secondary narrative; Sharp said to have left in 1968.	https://foreignpolicy.com/2017/08/25/the-secret-history-of-diplomats-and-invisible-weapons-russia-cuba/
15	Foster, K.R., Finch, E.D. — "Microwave hearing: evidence for thermoacoustic auditory stimulation by pulsed microwaves," Science (1974).	https://pubmed.ncbi.nlm.nih.gov/4833827/
16	Guy, A.W., Chou, C.K., Lin, J.C., Christensen, D. — "Microwave-induced acoustic effects in mammalian auditory systems and physical materials," Annals N.Y. Acad. Sci. (Feb. 28, 1975).	https://pubmed.ncbi.nlm.nih.gov/1054231/
17	Lin, J.C. — Microwave Auditory Effects and Applications (1978). Internet Archive scan.	https://archive.org/details/microwaveauditor0000linj
18	Foster, K.R., Garrett, J., Ziskin, M.C. — modern review of alleged microwave/RF 'weapon' claims; concludes practical weaponization with known systems is unlikely or operationally implausible.	https://pmc.ncbi.nlm.nih.gov/articles/PMC8733248/
19	WRAIR Finding Aid — Walter Reed Army Institute of Research archival collection, Series I/II.	https://wrair.health.mil/Portals/87/Documents/WRAIR%20Finding%20Aid.pdf?ver=GlZOh4AUYb5Jxl-qVWCtUA%3D%3D
20	WRAIR Library/Archival Collection — access procedures (in-person, U.S. citizens, by appointment; Gorgas Memorial Library, 503 Robert Grant Ave, Silver Spring, MD 20910).	https://wrair.health.mil/Library-Archival-Collection/
21	WRAIR FY1983 Annual Report (DTIC ADA147533) — Work Unit 214 Millimeter Wave Biophysics; Work Unit 041 Biological Interactions/Hazards of Microwave Radiation.	https://archive.org/stream/DTIC_ADA147533/DTIC_ADA147533_djvu.txt
22	"Services to Operate and Maintain a Microwave Research Laboratory," WRAIR/USAMRD (May 2004; DTIC ADA425649); contracts DAMD17-94-C-4069 / DAMD17-94-E-0001.	https://archive.org/stream/DTIC_ADA425649/DTIC_ADA425649_djvu.txt
23	PubMed — WRAIR Department of Microwave Research authorship continuity (1990s).	https://pubmed.ncbi.nlm.nih.gov/8285916/
24	PubMed — WRAIR Department of Microwave Research authorship continuity (supporting 1990s entry).	https://pubmed.ncbi.nlm.nih.gov/1447544/
25	U.S. Patent 6,470,214 B1 — "Method and device for implementing the radio frequency hearing effect," O'Loughlin & Loree; assignee U.S. Department of the Air Force; filed Dec. 13, 1996; issued Oct. 22, 2002 (expired).	https://patents.google.com/patent/US6470214B1/en

#	Source	URL
26	U.S. Patent 6,587,729 B2 — "Apparatus for audibly communicating speech using the radio frequency hearing effect" (divisional of US6470214); filed Apr. 24, 2002; issued July 1, 2003 (expired).	https://patents.google.com/patent/US6587729B2/en
27	GBPPR archived SBIR abstract — Phase I AF93-026, "Communicating Via the Microwave Auditory Effect," Brian Kohn, Science & Engineering Associates; Control 93AL-185; contract F41624-93-C-9013; $37,806; 17 May–17 Dec 1993. Mirror of a defunct federal abstract page.	http://gbppr.net/mil/mindcontrol/raven1/v2s-kohn.htm
28	USAF FOIA Log FY2000 (released, redacted) — case 00-F-0013 (Margo P. Cherney): "Copies of Communicating Via the Microwave Auditory Effect — DOD SBIR Contract #F41624-95-C-9007." Hosted by The Black Vault.	https://documents.theblackvault.com/documents/foia/FOIALog_FY2000redacted.pdf
29	U.S. Patent 4,877,027 — Brunkan, "Hearing system" (microwave-induced hearing using 100 MHz–10 GHz radiation, bursts of pulses, audio modulation). Patent claims, not deployment evidence.	https://patents.google.com/patent/US4877027A/en
30	New Scientist — "Microwave ray gun controls crowds with noise" (MEDUSA / Sierra Nevada). James Lin warns about power-level feasibility and possible neural damage.	https://www.newscientist.com/article/dn14250-microwave-ray-gun-controls-crowds-with-noise/
31	Wired — "The Other MEDUSA" (Aug. 2007): Phase I hardware design, experimental indications of microwave auditory effect; no field deployment documented.	https://www.wired.com/2007/08/the-other-medus/
32	National Academies of Sciences, Engineering, and Medicine — "An Assessment of Illness in U.S. Government Employees and Their Families at Overseas Embassies" (2020). Committee found acute signs/symptoms most consistent with directed, pulsed RF energy among the mechanisms considered, with limited information.	https://www.nationalacademies.org/read/25889
33	National Academies 2020 report — RF mechanisms chapter (directed, pulsed RF discussion).	https://www.nationalacademies.org/read/25889/chapter/11
34	Bartholomew, R.E., Baloh, R.W. — skeptical review of Havana-syndrome literature; cites 2023 U.S. intelligence-community assessment that a foreign sonic or microwave device is 'highly unlikely' to explain reported cases.	https://pmc.ncbi.nlm.nih.gov/articles/PMC10913303/
35	Yaghmazadeh et al. — review of high-power pulsed RF bioeffects literature; biological plausibility for some neurological effects, but does not validate operational voice-to-skull speech communication.	https://pmc.ncbi.nlm.nih.gov/articles/PMC10508825/
36	Justesen, D.R. — "Behavioral and psychological effects of microwave radiation," Bull. NY Acad. Med. (1979). PMC archive. Does not repeat the Sharp/Grove voice/digit anecdote.	https://pmc.ncbi.nlm.nih.gov/articles/PMC1807730/
37	WRAIR annual report (DTIC ADA009338) — Walter Reed Army Institute of Research Annual Progress Report covering 1973–1974 microwave research environment; no matching voice/digit experiment in the public record.	https://archive.org/details/DTIC_ADA009338

End of memorandum. Prepared by Perplexity Computer on May 8, 2026. Working draft for counsel review. Not legal advice.